Advances in Finite Element Methods and Parallel Scientific Computing

November 28 – December 2, 2026
IIT Bhubaneswar

Academic Program
Venue: Room No. L02M, Lalitgiri Hall of Learning

Inauguration Program: 9:00 AM – 9:30 AM, L02M (Lalitgiri Hall of Learning), 28th November 2026

Day 1: November 28, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Bhupen Deka
9:30 AM – 10:10 AM	**Prof. Rajen Kumar Sinha** *A geometric transformation based finite element method for the elliptic interface problems*
10:15 AM – 11:05 AM	**Dr. Asha Dond** *Lecture Series on Adaptive FEM: Convergence and Quasi-optimality – Lecture 1*
11:05 AM – 11:35 AM: **TEA BREAK**	
11:35 AM – 12:25 PM	**Dr. Pratibhamoy Das** *Lecture Series Parallel Computing – Lecture 1*
12:30 PM – 12:50 PM	**Contributed Talk: Rohit Kumar Yadav** *A parameter uniform Numerical method for Singularly Perturbed Semilinear Convection diffusion problems with discontinuous source term*
1:00 PM – 2:30 PM: **LUNCH BREAK**	
	Chair: Dr. Pratibhamoy Das
2:30 PM – 3:10 PM	**Dr. Sangita Yadav** *Hybridizable Discontinuous Galerkin (HDG) Method*
3:15 PM – 3:35 PM	**Contributed Talk: Ajay Kumar** *Hybrid high-order methods for sixth order problems with a mixed formulation*
3:40 PM – 4:00 PM	**Contributed Talk: Suraj Kumar** *Parameter-Uniform Convergence and Superconvergence of NIPG Method for Degenerate Singularly Perturbed Convection–Diffusion Problems*
4:00 PM – 4:25 PM: **TEA BREAK**	
4:25 PM – 4:45 PM	**Contributed Talk: Dilip Sarkar** *A Finite Element Framework for Domain Decomposition with Non-Matching Grids: A Nitsche-Based Approach*
4:50 PM – 5:10 PM	**Contributed Talk: Kumar Mritunjay** *Boundary Feedback Stabilization and Finite Element Approximation of a Strongly Damped Semilinear Wave Equation*

Day 2: November 29, 2026

Time	Speaker / Title of the Talk
	Chair: Dr. Bankim Chandra Mandal
9:30 AM – 10:20 AM	**Dr. Pratibhamoy Das** *Lecture Series on Parallel Computing - 2*
10:25 AM – 11:05 AM	**Prof. Neela Natarajan** *Unified numerical analysis for thermoelastic diffusion and thermo-poroelasticity of thin plates*
11:05 AM – 11:35 AM: **TEA BREAK**	
11:35 AM – 12:15 PM	**Prof. Amiya K. Pani** *On Backward Time-Fractional Diffusion Problems: A Unified Approach*
12:20 PM – 12:40 PM	**Contributed Talk: Dr. Gouranga Pradhan** *Virtual Element Methods for the General Linear Second-order Hyperbolic Problems*
12:40 PM – 1:00 PM	**Contributed Talk: Dr. Ritesh Kumar Singh** *Exact Solution of the Riemann Problem in a Perfect Gas with Magnetic Field Using Lie Symmetry Analysis*
1:00 PM – 2:30 PM: **LUNCH BREAK**	
	Chair: Dr. Sangita Yadav
2:30 PM – 3:20 PM	**Dr. Arbaz Khan** *Lecture Series on Mixed FEM for Multiphysics Problems - 1*
3:25 PM – 3:45 PM	**Contributed Talk: Rakesh Sahoo** *Parareal Methods for Vlasov Systems in Plasma Physics*
3:50 PM – 4:15 PM: **TEA BREAK**	
4:20 PM – 4:40 PM	**Contributed Talk: Arijit Pal** *Dimensional-Splitting Weak Galerkin FEM for Singularly Perturbed Initial-Boundary Value Problems*
4:40 PM – 5:00 PM	**Contributed Talk: Dr. Rashmita Baruah** *A Weak Galerkin Method with Forward Euler Time Discretization for Singularly Perturbed Parabolic Interface Problems*
5:00 PM – 5:20 PM	**Contributed Talk: Pratyay Mondal** *Stability and convergence of a decoupled discontinuous Galerkin scheme for the Oldroyd Model of order one*

Day 3: November 30, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Rajen Sinha
9:30 AM – 10:20 AM	**Dr. Asha Dond** *Lecture Series on Adaptive FEM: Convergence and Quasi-optimality – Lecture 2*
10:25 AM – 11:05 AM	**Prof. Rajendra K. Ray** *New Higher-Order Super-Compact Schemes for Three-Dimensional Fluid Flow and Heat Transfer in Non-Newtonian Fluids*
11:05 AM – 11:25 AM: **TEA BREAK**	
11:25 AM – 12:15 PM	**Prof. Bhupen Deka** *Lecture series on Weak Galerkin Solver for Elliptic BVP on Polygonal Meshes - 1*
12:20 PM – 1:00 PM	**Dr. Y. Bhumkar** *Scientific Computing of Computational Aero Acoustic Problems*
1:00 PM – 2:30 PM: **CONFERENCE LUNCH**	
3:00 PM – 7:30 PM	**Free Time/Local Excursion**

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Day 4: December 1, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Rajendra K. Ray
9:30 AM – 10:20 AM	**Dr. Arbaz Khan** *Lecture Series on Mixed FEM for Multiphysics Problems – Lecture 2*
10:25 AM – 11:05 AM	**Dr. Rakesh Kumar** *Divergence-free reconstruction of 2-D vector fields: Application to induction equation*
11:05 AM – 11:15 AM: **TEA BREAK**	
11:20 AM – 12:00 PM	**Dr. Nagaiah Chamakuri** *Computational Challenges in Cardiac Electromechanics: Robust Algorithms and Scalable Solvers*
12:05 PM – 12:25 PM	**Contributed Talk: Deeksha Tomer** *Domain Decomposition Methods for PDEs with Time Delay*
12:25 PM – 1:05 PM	**Contributed Talk: Jitendra Nath Naik** *Optimal Error Estimates of a Finite Element Method for Semilinear SPDEs with Non-smooth Initial Data*
1:10 PM – 2:30 PM: **LUNCH BREAK**	
	Chair: Dr. Sanjib Acharya
2:30 PM – 3:20 PM	**Prof. Bhupen Deka** *Lecture series on Weak Galerkin Solver for Elliptic BVP on Polygonal Meshes - 2*
3:20 PM – 3:40 PM	**Contributed Talk: Monirul Islam** *Unconditionally Energy Stable Ultra-Weak Discontinuous Galerkin Scheme with Generalized Numerical Fluxes for the Allen-Cahn Equation*
3:40 PM – 4:20 PM	**Dr. Surya Maharana** *On Applications of FEM for the Study of Chemo-hydrodynamic Instabilities*
4:20 PM – 4:35 PM: **TEA BREAK**	
4:35 PM – 4:55 PM	**Contributed Talk: Ashish Kumar Saini** *Enhancing thermal performance in square enclosures filled with oxide nano-fluids subjected to magnetic fields: a numerical study*
4:55 PM – 5:15 PM	**Contributed Talk: Dr. Harish Mallesham Nagula** *Vectorized Mortar Primal Hybrid Finite Element Method*

Day 5: December 2, 2026

Time	Speaker / Title of the Talk
	Chair: Dr. Nagaiah Chamakuri
9:30 AM – 10:20 AM	**Prof. Sarvesh Kumar** *Virtual Element Methods: Recent Developments and Applications*
10:25 AM – 11:05 AM	**Dr. Bikram Bir** *Discontinuous Galerkin Finite Element Methods for the Electro-hydrodynamic Problems*
11:05 AM – 11:30 AM: **TEA BREAK**	
11:30 AM – 11:50 AM	**Contributed Talk: Dr. Achyuta Ranjan Dutta Mohapatra** *Energy norm error estimates of a hybrid high-order method for the linear parabolic integro-differential equations on general meshes*
11:55 AM – 12:15 PM	**Contributed Talk: Anupama Ghorai** *A high-order finite element method for multi-dimensional population balance equations.*
12:20 PM – 12:40 PM	**Contributed Talk: Ayush Agrawal** *Nitsche method for the Stokes-Poisson-Boltzmann equation with Navier slip boundary condition.*
12:45 PM – 1.05 PM	**Contributed Talk: Dr. Arya Prakash Padhi** *Scalable One-Shot 3D Topology Optimization with Multigrid PCG and GPU-Accelerated Finite Element Analysis*
1:00 PM – 2:30 PM: **LUNCH BREAK**	
	Chair: Dr. Arbaz Khan
2:35 PM – 2:55 PM	**Contributed Talk: Dr. Aniruddha Seal** *C^0 interior penalty method for time-fractional Cahn-Hilliard equation*
3:00 PM – 3:20 PM	**Contributed Talk: Dr. Ruma Rani Maity** *An Energy-Decreasing Algorithm for the Finite Element Approximation of Ferronematic Equilibrium States*
3:25 PM – 3:45 PM	**Contributed Talk: Dr. Shubhangi Dwivedi** *Managing Connected Ecological Patches Under Species Dispersal Using Optimal Control*
3:50 PM – 4:30 PM: **CLOSING CEREMONY**	